Pre-1997 FreeLIP validation

Fixed NIST/ANSI cohort | 27 October 2026 | T-5518

The missing historical selector remains unresolved. This observation report records a recovered older library and completed tests. Every tested validation operation passed on all 11 reconstructed omissions and all 10 publications, across all six streams. No formal hypothesis status changes here.

A dated source gap is now smaller

Two mirrors supplied the same FreeLIP 1.0 archive: 201,037 bytes, README dated May 1, 1995, HTTP modification date April 30, 1996. The previously tested 1.1 README is dated July 21, 1997 and describes a fixed extended-Euclidean bug that could produce wrong answers, hangs or memory errors. The recovered source is not authenticated as Reiter's generator.

Recorded observations

Configuration	Inverse / alias checks	May base points
FreeLIP 1.0 / 26 bits	189 / 189 pass	6 / 6 match
FreeLIP 1.0 / 30 bits	189 / 189 pass	0 / 6 match
FreeLIP 1.1 / 26 bits	189 / 189 pass	not rerun
FreeLIP 1.1 / 30 bits	189 / 189 pass	not rerun

FreeLIP 1.0 coefficient inversion, square root and four-seed order-primality checks pass on all 21 curves in both limb sizes. The A/A repeats match byte for byte. There are 168 distinct primality parameter combinations, 336 calls including repeats. Independent Python arithmetic verifies 126 returned inverse integers.

Unique curves passing every applicable check

Units: unique curves. n = 21 fixed identities, six streams; deterministic counts, no error bars.

Figure 1. Recorded unique-curve outcomes. FREELIP-PRE1997-VALIDATION-02 and WITNESS-AND-FAULT-CONTROL-01. Full identities and exact arithmetic inputs are in comparison.csv.

Methods, controls and preserved failures

What actually ran

Each curve contributes three inverse inputs: c, nonzero b, and the j-denominator (27 b squared - 108) mod p. Each is tested with distinct output, aliased output, and pre-sized aliased output, in both limb sizes and both source versions: 21 x 3 x 3 x 2 x 2 = 756 calls. All pass. The 126 independent integer witnesses cover FreeLIP 1.0, three inputs, and both limb sizes.

Ordinary controls pass in all four builds: 68 prime/composite cases, four square-root controls and 40 independently replayed RNG words. In 1.0, coefficient square roots consume zero FreeLIP random calls, and each 20-round primality call consumes 20. Both preserve the separate libc rand48 stream. Seven discarded 26-bit draws reproduce all six May base-point x-coordinates. The 30-bit mismatch identifies a configuration difference.

The constructed memory-fault demonstration is unavailable

Attempt 01 stopped at its first compile because the archive expects a generated lippar.h. No curve ran. The source Makefile creates an empty bootstrap header; the frozen attempt-02 amendment supplies it, with arithmetic options explicit on the compiler command line.

The first ASan control timed out without producing probe output. A second was sampled in loader / libSystem / sanitizer initialization, recursively entering its allocator and spinning in an initialization mutex, with no main frame. It was interrupted; its stdin and missing terminal capture are recorded. Remaining ASan arms did not start. This is pre-main operational censoring, not a demonstrated FreeLIP bug or a curve rejection. All 126 independent witnesses had already completed.

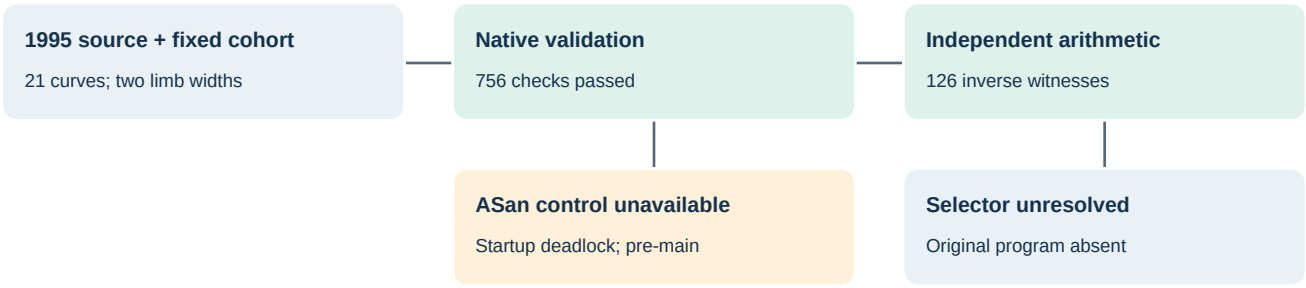

Figure 2. Completed validation and independent replay are separate from the unavailable constructed fault control and unresolved historical attribution. Editable diagram data and SVG accompany this report.

Scope of the source comparison

Top-level square-root, primality and RNG routine bodies match after whitespace/comment removal. Extended Euclid, prime iteration and other helper differences required native testing. Both tested C versions use modern macOS ARM64/LP64, SIZEOFLONG=8, explicit 26/30-bit limbs and generic arithmetic. These observations do not establish original-CPU behavior, allocator behavior, original timing or an application-level selector.

Frozen native protocol: f246a5a499f7b6d21273e911bbe509f117f645b26b629500d76f8169145fa9a9

Historical implications and replay

No recovered cube-root gate

The 1994 Lehmann-Maurer-Mueller-Shoup paper constructs eta-quotient modular equations directly from j(E) (page 5). Its presented path does not require a cube root of j in the base field. Special j-values have separate handling (page 4); Frobenius checks include x and y (page 6); fingerprint matches are followed by full-point comparison (page 9). All ten pages were read visually; OCR covers pages 1-7 only.

The recovered 1995 MIRACL package contains arithmetic/factoring software; the available prime-field Schoof/SEA source dates from 1999. The 1995/1996 elliptic packages inspected concern characteristic two. LiDIA-history and its preserved Software Heritage snapshots have empty branch maps. No older implementation of the later LiDIA fixes was recovered. These were bounded searches; access timeouts do not prove historical source absence.

The P-224 cube/splitting pattern remains a specific field-dependent lead. This round provides no source-based cube-root gate and no selector that rejects the eleven omissions while retaining all publications. Another point counter, application stopping logic, or an unrecorded criterion remains possible.

Exact identity and custody

Fixed cohort: 21 identities, ten publications, eleven reconstructed omissions, six streams. Full UIDs, field p, order n, c, b, timestamps and ordinals are retained in comparison.csv and SUMMARY.json. The model is y squared = x cubed - 3x + b. Published 192v2 is ordinal 6013 and 192v3 is 5228; P-224 coefficient signs represent one class in this study.

Cohort SHA-256: 93ac4774b993a994a17d9327df92e0f2b287e66755cd933fdc683015fec1d545

Archive: 190,218 bytes; 3,688 sealed files; 912 completed external-process receipts, including the compile failure and sanitizer timeout. Original library source and binaries remain in local custody; the public derived bundle contains probes, inputs, metadata, raw observations and portable auditors.

Archive SHA-256: a7d8001e049cf0a312eba975710a226f00e34d2abc66f83aa9b31b3a03bc7972

Portable verification

Extract evidence.tar.zst. Run `python3 -B evidence/verify.py`, then `python3 -B audit_bindings.py evidence`. Both passed after relocation. The first checks the sealed files, exact cohort, ordinary controls, arithmetic witnesses and recorded unavailable control. The second binds all summary rows to raw stdin, stdout and process receipts. Neither launches native libraries or accesses the network.

Earlier SEA and ECPP graphs retain their original sealed-run scope; their observations did not change. This round supplies separate library plots. Further historical attribution still requires the original generator, a dated relevant point counter, or a documented acceptance condition that survives the full cohort comparison.

Primary sources: ftp.funet.fi/pub/crypt/math/freelip_1.0.tar.gz; matching ftp.gwdg.de/pub/misc/crypt/math/freelip_1.0.tar.gz; github.com/lrobot/freelip (1.1); www.shoup.net/papers/counting.pdf (1994). Exact URLs and retrieval/source digests are in SOURCE.json and SEARCH-SOURCES.json.